Mira Risk Engine

Official report: detect risk, size it, act. Backtest 2001-2026 and strict 2025-2026 test.
Version 1.0, 2026-10-15. Project: Mira (U-Hackathon Flow Summit). Research only. No real money. Not investment advice.

> **Read first.** This is a historical backtest with a fixed rule set. It is not a forecast and not a promise of return. Markets can fall further than any past period. For real money, speak to a licensed adviser.

1. Summary

-8%
worst fall, Mira chain 60%, vs -33% fixed and -51% all-stocks

+5.9%
growth a year, Mira chain 60% (fixed +6.2%, all-stocks +8.8%)

0.76
Sharpe over cash (fixed 0.52)

5 of 26
calendar years the chain beat fixed 60%

The engine watches the S&P 500 for three danger signs, estimates how bad one month could be, and lowers the stock share when risk is high. Over 25 years it kept a growth rate close to a fixed 60% stock mix while avoiding the deepest falls. In the calm 2025-2026 window it gave up growth because there was little to protect against.

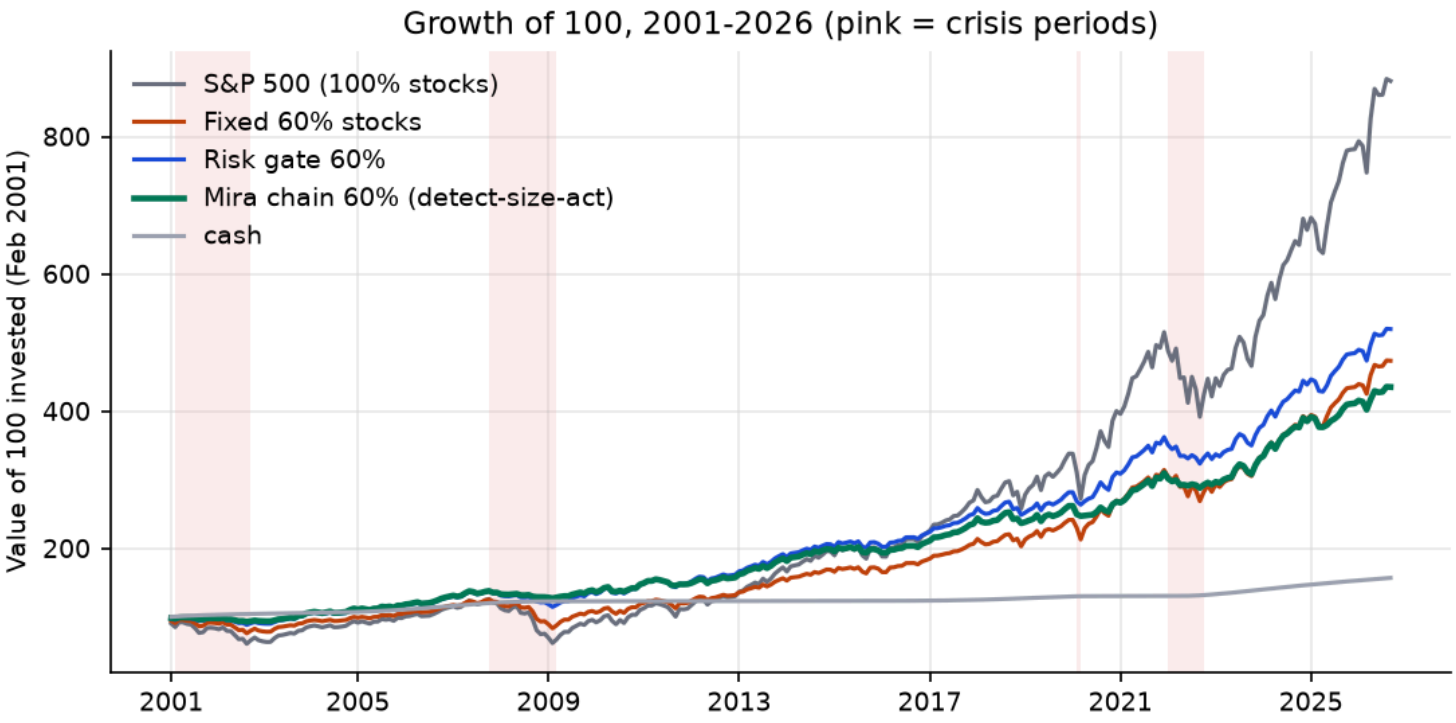
Best points: shallow falls in the dot-com bust, the 2008 crisis and Covid; a high Sharpe ratio; it works with a plain cash fallback; the rules are simple enough to explain to a first-time investor.
Weak points: it sometimes cuts stocks just before a rebound (April 2025, March 2026); it cannot see a sudden fall that starts from a calm market; its edge is avoiding deep falls, not extra return.

2. How it works

Step	What it does	Rule (fixed before testing)
1 Detect	Three alarms on the S&P 500 at month end	20-day volatility > 1.5 x 250-day; drawdown ≥ 10% from the 1-year high; price below the 200-day average. Stress = number of alarms (0-3)
2 Size	One bad month at 95% confidence	1.645 x the higher of 20/60-day volatility x √21. Stock share allowed by a 4% monthly risk budget
3 Act	Stock share = lower of base dial and risk-budget share, times a gate	Gate by stress: 0 = 100%, 1 = 75%, 2 = 50%, 3 = 25%. The rest sits in cash (long test) or a defensive multi-asset core

Costs: 10 basis points on every change. Decisions at month end use only data known then; money works the next month. Leak test: 12 of 12 random months gave the identical signal with future prices removed.

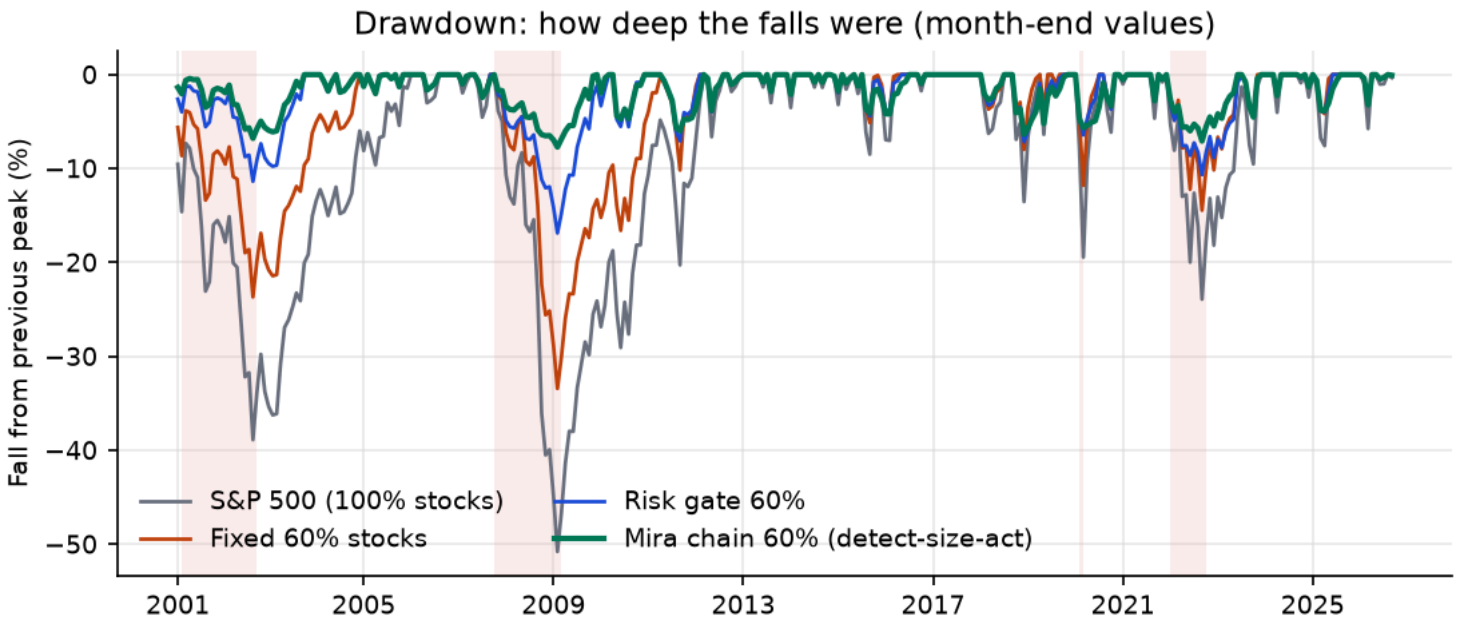
3. Results, 2001-2026 (S&P 500 total return + cash)



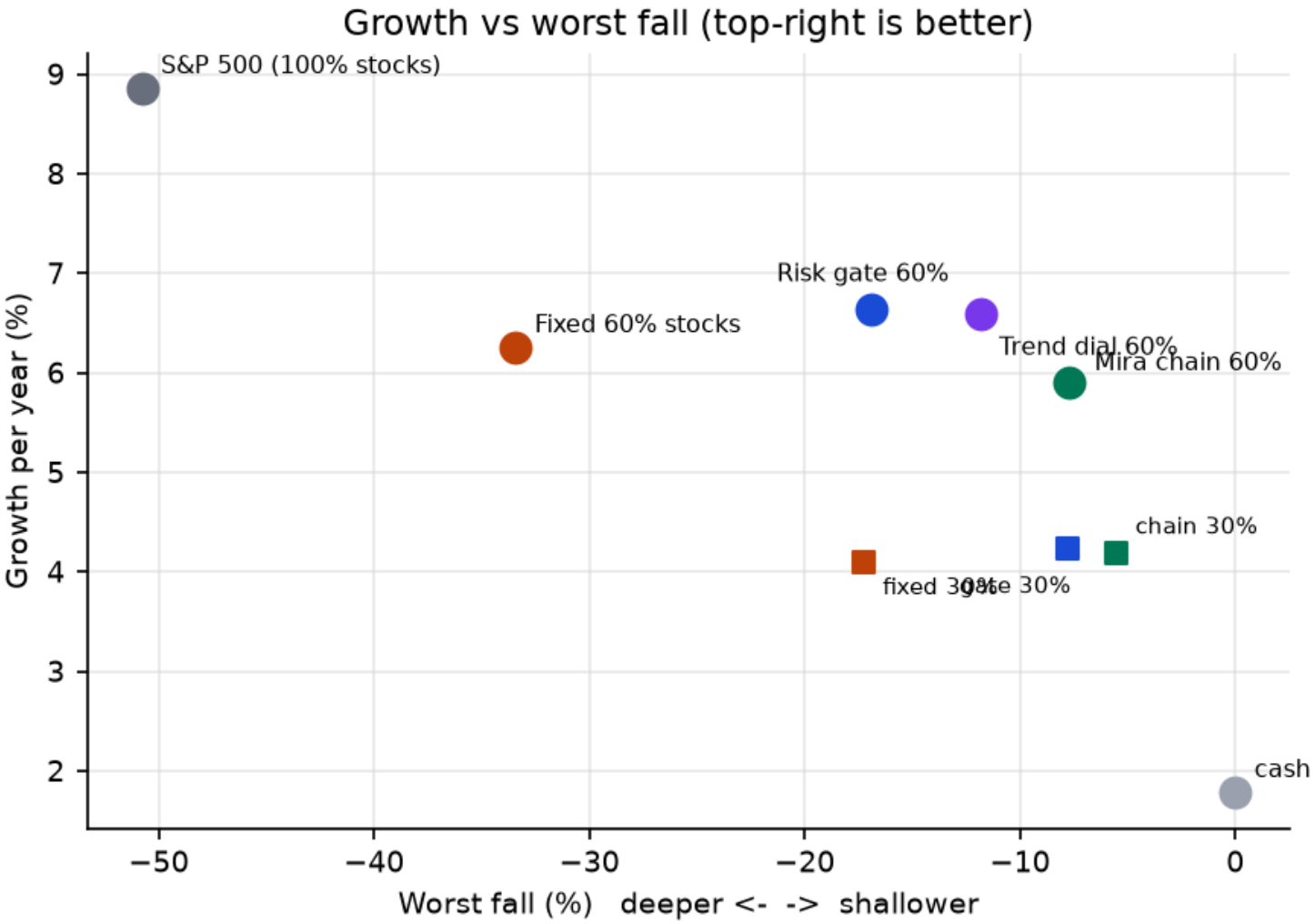
Growth of 100. The chain (green) stays close to the fixed mix and avoids the deepest falls.

60% base	Growth a year	Volatility	Sharpe	Worst fall	Worst 12 months	Up months	100 becomes
S&P 500 (100% stocks)	+8.8%	15.1%	0.52	-50.8%	-43.4%	65%	881
Fixed 60% stocks	+6.2%	9.0%	0.52	-33.4%	-28.0%	66%	474
Trend dial 60%	+6.6%	6.6%	0.74	-11.9%	-8.4%	75%	515
Risk gate 60%	+6.6%	6.5%	0.75	-16.9%	-12.0%	66%	520
Mira chain 60% (detect-size-act)	+5.9%	5.5%	0.76	-7.7%	-6.4%	66%	435
cash	+1.8%	0.5%	nan	+0.0%	+0.0%	100%	157

30% base	Growth a year	Volatility	Sharpe	Worst fall	Worst 12 months	Up months	100 becomes
Fixed 30% stocks	+4.1%	4.5%	0.52	-17.3%	-14.4%	66%	280
Risk gate 30%	+4.2%	3.3%	0.74	-7.8%	-5.6%	68%	290
Mira chain 30% (detect-size-act)	+4.2%	3.2%	0.75	-5.5%	-4.2%	68%	287

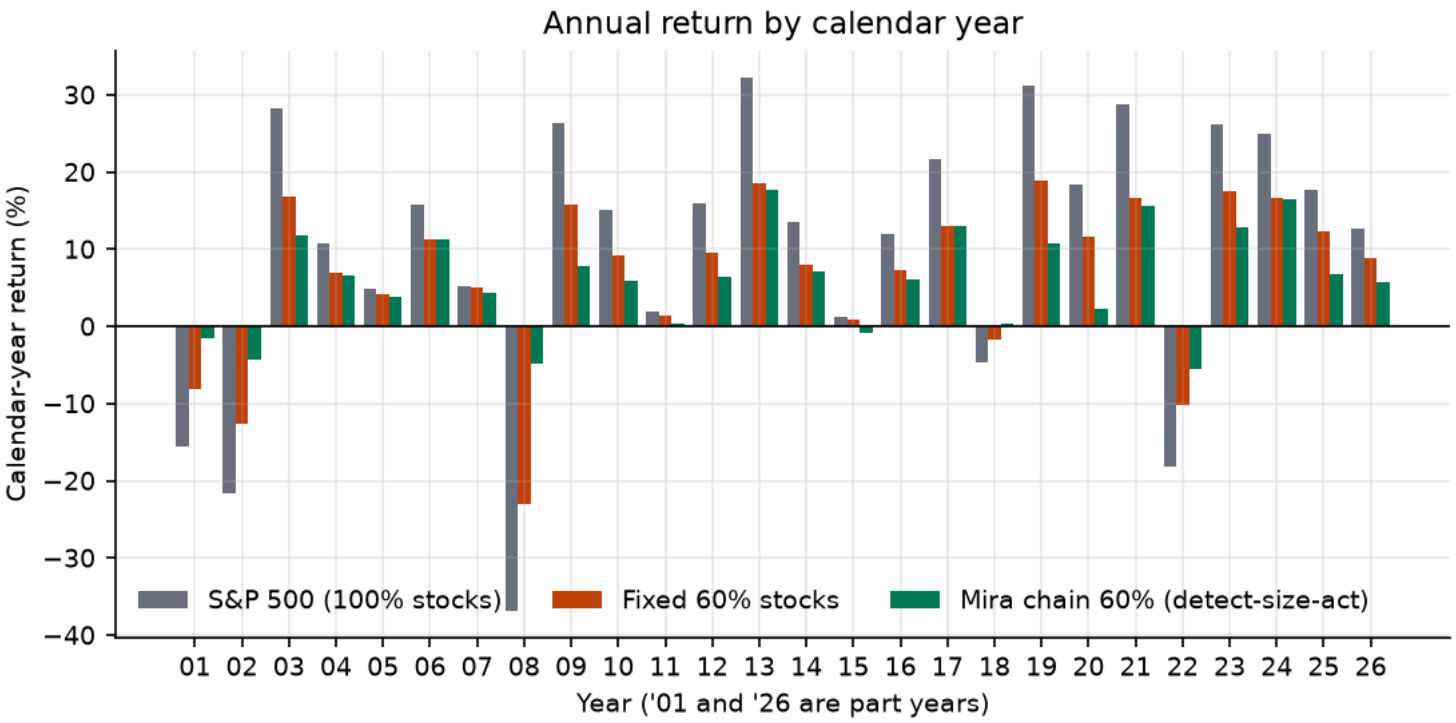


Drawdown. The chain falls far less than all-stocks or fixed 60% in every crisis.



Growth against worst fall. Top-right is better.

4. Returns by year and by month

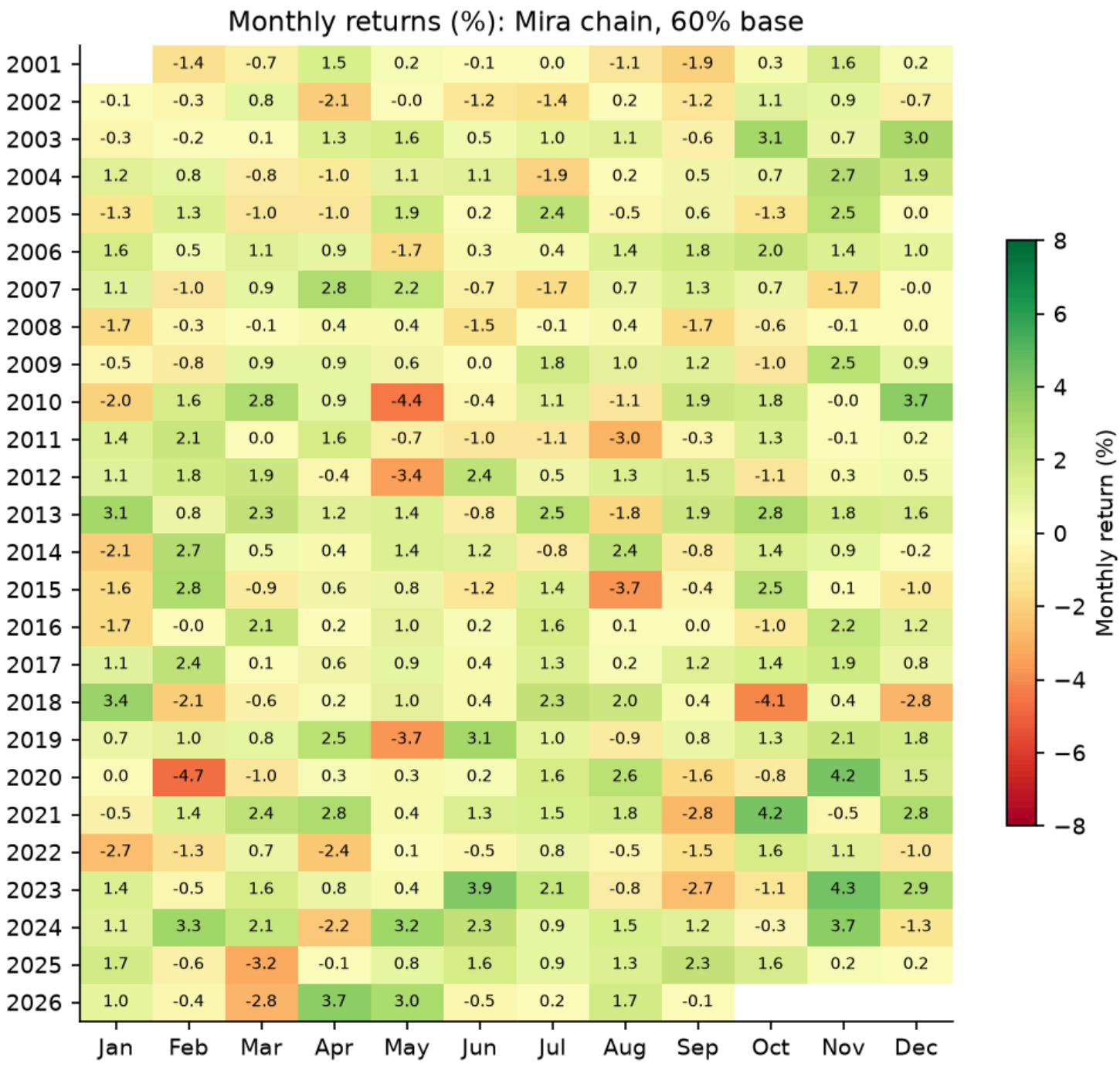


Annual return by calendar year: all-stocks, fixed 60% and Mira chain 60%.

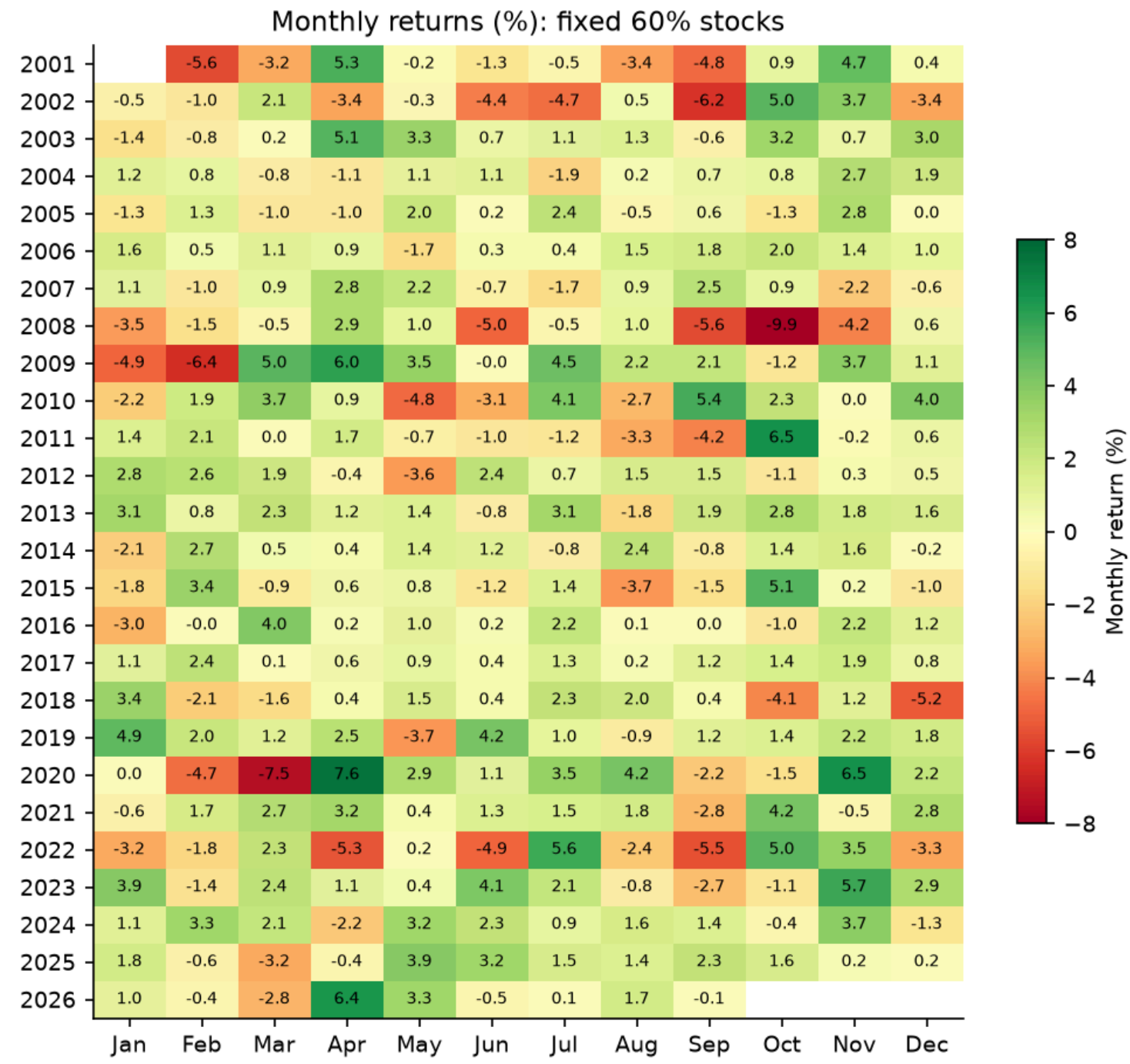
Year	S&P 500	Fixed 60%	Gate 60%	Mira chain 60%
2001 (Feb-Dec)	-15.5%	-8.1%	-2.5%	-1.5%
2002	-21.6%	-12.6%	-6.5%	-4.2%
2003	+28.2%	+16.8%	+14.4%	+11.8%
2004	+10.7%	+6.9%	+6.5%	+6.7%
2005	+4.8%	+4.2%	+4.2%	+3.8%
2006	+15.8%	+11.3%	+11.3%	+11.3%
2007	+5.1%	+5.0%	+4.7%	+4.4%
2008	-36.8%	-23.0%	-9.8%	-4.9%
2009	+26.4%	+15.8%	+12.2%	+7.8%
2010	+15.1%	+9.3%	+7.6%	+6.0%
2011	+1.9%	+1.4%	+1.3%	+0.3%
2012	+16.0%	+9.5%	+8.7%	+6.5%
2013	+32.3%	+18.5%	+18.5%	+17.7%
2014	+13.5%	+8.0%	+7.5%	+7.2%
2015	+1.2%	+0.9%	+0.4%	-0.9%

2016	+12.0%	+7.3%	+7.1%	+6.0%
2017	+21.7%	+13.0%	+13.0%	+13.0%
2018	-4.6%	-1.8%	-0.6%	+0.3%
2019	+31.2%	+19.0%	+13.2%	+10.7%
2020	+18.3%	+11.6%	+10.3%	+2.3%
2021	+28.7%	+16.6%	+16.6%	+15.6%
2022	-18.2%	-10.2%	-8.8%	-5.5%
2023	+26.2%	+17.6%	+13.8%	+12.9%
2024	+24.9%	+16.7%	+16.7%	+16.5%
2025	+17.7%	+12.2%	+10.5%	+6.8%
2026 (Jan-Sep)	+12.7%	+8.8%	+7.2%	+5.8%

Worst calendar year: all-stocks -36.8%, Mira chain -5.5%.

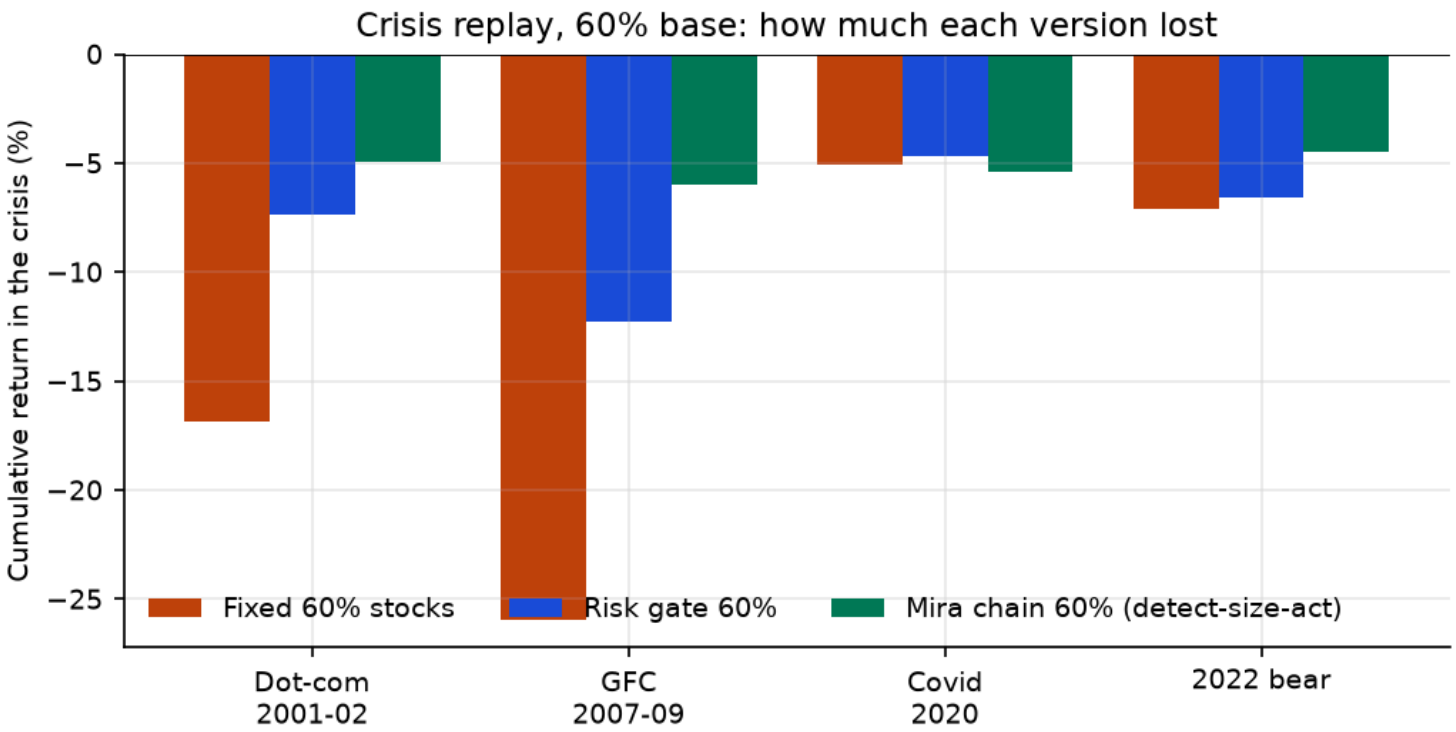


Month-by-month returns of the Mira chain (60% base). Green = gain, red = loss.



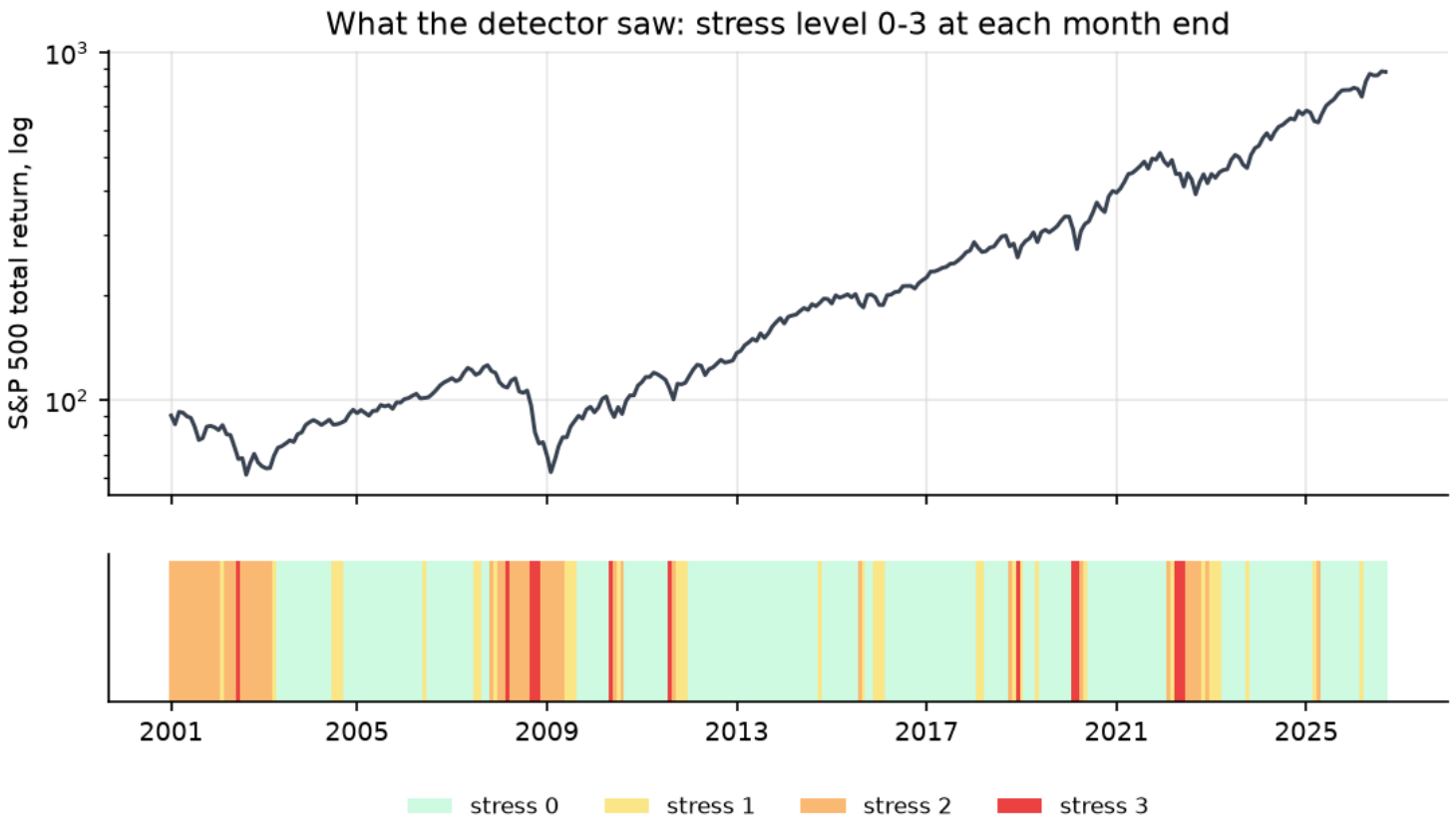
Same view for the fixed 60% mix, for comparison.

5. Crises and what the detector saw



Cumulative return in each crisis, 60% base.

Crisis, cumulative return	Fixed 60%	Gate 60%	Mira chain 60%
Dot-com 2001-01..2002-10	-16.9%	-7.3%	-5.0%
GFC 2007-10..2009-03	-25.9%	-12.2%	-6.0%
Covid 2020-01..03	-5.1%	-4.7%	-5.4%
2022 bear 2021-12..2022-10	-7.1%	-6.6%	-4.5%



Stress level (0-3) at each month end under the S&P 500 total return. Darker = more alarms.

Stress 2 or 3 came before 11 of the 13 one-month falls of 8% or more. After stress 3, the next month fell 5% or more in 46% of months (13 months, a small sample). Stress is a volatility-regime gauge, not a crash forecast: falls that start from a calm market are not seen in time.

6. Strict test, 2025-2026

Decisions from January 2025, after the training cutoff of the AI model used elsewhere in the project. Signals were checked for look-ahead. Defensive side: the multi-asset core. 19 months.

60% base	Growth a year	Worst fall	Sharpe
Fixed 60% stocks	+12.5%	-4.0%	1.03
Risk gate 60%	+10.4%	-3.9%	0.90
Mira chain 60% (detect-size-act)	+7.5%	-3.8%	0.59

The market was calm; the worst fall for the fixed mix was about -4%. The declared success rule is inconclusive. The gate lowered growth because it cut stocks in April 2025 and March 2026, before rebounds.

7. Honest limits

- The rules were fixed before the runs, but the authors saw the 2008-2026 results. Only the forward test is unseen.
- 2017-2026: the chain did not meet the declared success rule (stocks mostly rose; protection cost growth).

- Few independent crises (about four). Intervals on return differences include zero: the benefit is shallower falls, not extra return.
- The long test uses plain cash as the defensive side and the S&P 500 only; taxes and fees beyond 10 bps are not modelled.
- The AI news layer did not beat the statistical anchor in the sealed 2025-2026 test, so its weight in the system is 10%.

8. Forward test (pre-registered)

The rule was frozen on 2026-10-15 with a code hash and a monthly ledger. Decisions logged so far: 1. A verdict needs 12 scored months and starts from October 2026 returns. A code change makes the program refuse to run.

9. Reproduce

Code and data snapshot: `risk-system_SNAPSHOT_2026-10-15_mira.zip`. Commands: `python -E mira_invest.py`, `python -E mira_sandbox_tests.py`, `python -E mira_official_report.py`. Tests: 19 suites, all passing.